The Pigeonhole Principle in Self-Attention: Rank Barriers and Collision Limits

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Abstract

We prove three fundamental limitations on the expressivity of single-head self-attention at random initialization. First, we establish a **rank barrier**: the attention score matrix $Z = QK^T/\sqrt{d_k}$ has rank at most d_k, preventing any configuration from producing full-rank attention patterns when the sequence length n exceeds d_k. Second, we prove a **collision theorem**: under Xavier initialization, the expected fraction of positions experiencing a near-tie in their attention scores grows with n and remains bounded away from zero for fixed d_k. Third, we bound the **expected selectivity** at initialization as $\mathbb{E}[\text{SEL}] \le C \cdot d_k/n$, showing that average maximum attention weight vanishes as $O(d_k/n)$ when d_k is fixed and $n \to \infty$. All three theorems are empirically verified on NVIDIA Jetson Orin GPU via bit-exact PyTorch simulations spanning 500 random initializations across multiple (n, d, d_k) configurations. Our results place hard structural and probabilistic limits on what attention can represent without learning, with implications for model initialization theory and architectural design.

1. Introduction

The Pigeonhole Principle states that if n items are placed into m containers with $n > m$, at least one container must contain more than one item. In the context of transformer self-attention (Vaswani et al., 2017), the "items" are the n positions in a sequence, and the "containers" are the d_k-dimensional key space into which each position projects.

When $n > d_k$, the Pigeonhole Principle applies directly: n query vectors and n key vectors all live in $\mathbb{R}^{d_k}$, forcing structural dependencies. We show that these dependencies translate into three concrete limitations on attention expressivity at initialization, before any gradient-based learning has occurred.

1.1 Contributions

1. **Rank Barrier (Theorem 1):** The score matrix Z has rank at most d_k. When $n > d_k$, full-rank target patterns (e.g., permutation attention) are structurally unattainable regardless of weight values.

2. **Collision Theorem (Theorem 2):** Under Xavier initialization, the expected collision fraction satisfies $\mathbb{E}[\text{collisions}/n] \geq 1 - \exp(-\delta^2 d_k / C)$, remaining bounded away from zero for fixed d_k as $n \to \infty$.
3. **Selectivity Vanishing (Theorem 3):** The expected selectivity $\mathbb{E}[\text{SEL}] \leq C \cdot d_k/n$, proving that average maximum attention weight must shrink as sequence length grows.

1.2 Related Work

Bhattamishra et al. (2020) established that transformers can simulate Turing machines given sufficient depth and width, but their construction assumes d_k scales with n. Our results show that when d_k is fixed (the practical regime), single-head attention is structurally limited even before learning begins.

2. Preliminaries

2.1 Self-Attention Architecture

For input representations $X \in \mathbb{R}^{n \times d}$, single-head attention computes:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$Z = \frac{QK^T}{\sqrt{d_k}} \in \mathbb{R}^{n \times n}$$

$$A = \text{softmax}(Z) \in \mathbb{R}^{n \times n}$$

where $W_Q, W_K \in \mathbb{R}^{d \times d_k}$ and softmax is applied row-wise.

2.2 Definitions

Definition 1 (Collision). For row i of score matrix Z and threshold $\delta \in (0,1)$, a **collision** occurs if:

$$\max_j Z_{ij} - \max_{j \neq j^*} Z_{ij} \ \leq \ \delta \cdot \max_j |Z_{ij}|$$

where $j^* = \arg\max_j Z_{ij}$.

Definition 2 (Selectivity). The **selectivity** of attention matrix A is:

$$\text{SEL}(A) \ = \ \frac{1}{n} \sum_{i=1}^{n} \max_j A_{ij}$$

3. Rank Barrier

Lemma 1 (Rank of Score Matrix). Let $Q, K \in \mathbb{R}^{n \times d_k}$. Then $\text{rank}(QK^T) \leq \min(\text{rank}(Q), \text{rank}(K)) \leq d_k$.

Proof. The product of an $n \times d_k$ matrix and a $d_k \times n$ matrix has rank at most the minimum of the inner dimension. Since Q and K each have d_k columns, $\text{rank}(Q) \leq d_k$ and $\text{rank}(K) \leq d_k$. The product bound follows from submultiplicativity. ■

Theorem 1 (Rank Barrier). For any single-head attention with query/key dimension d_k and any input $X \in \mathbb{R}^{n\times d}$, the attention score matrix $Z = XW_Q W_K^T X^T/\sqrt{d_k}$ has rank at most d_k.

Consequently, if $n > d_k$, there exists no weight configuration producing a score matrix with rank n. In particular, any target attention pattern requiring n linearly independent score rows is unattainable.

Proof. Since $Q = XW_Q$ and $K = XW_K$, we have $Z = QK^T/\sqrt{d_k}$. By Lemma 1, $\text{rank}(Z) \leq d_k$. If $n > d_k$, Z cannot have full rank n. Any attention pattern requiring n independent score rows (e.g., a permutation matrix with rank n) is therefore unattainable regardless of weight values. ■

Corollary 1 (Permutation Attention Impossibility). A single-head attention layer with $d_k < n$ cannot produce a permutation matrix P_π as its attention pattern for any non-trivial permutation π.

Proof. A permutation matrix has rank n (all singular values equal 1). By Theorem 1, $\text{rank}(A) \leq d_k + 1 < n$. Therefore $A \neq P_\pi$ for any permutation π. ■

4. Collision at Initialization

Lemma 2 (Gaussian Score Distribution). Under Xavier initialization ($W_Q, W_K \sim \mathcal{N}(0, I_d/d)$) and isotropic input X with rows $x_i \sim \mathcal{N}(0, I_d)$, the score $z_{ij} = q_i^T k_j/\sqrt{d_k}$ conditioned on q_i is distributed as $\mathcal{N}(0, \|q_i\|^2/d_k)$.

Proof. Since k_j is isotropic Gaussian, the linear form $q_i^T k_j$ is Gaussian with variance $q_i^T q_i = \|q_i\|^2$. Scaling by $1/\sqrt{d_k}$ gives the stated variance. ■

Theorem 2 (Collision Theorem). Under Xavier initialization, for any $\delta \in (0, 1)$, the expected collision fraction satisfies:

$$\mathbb{E}\left[\frac{\#\text{collisions}}{n}\right] \;\geq\; 1 - \exp\left(-\frac{\delta^2 d_k}{C}\right) - O\left(\frac{1}{\sqrt{n}}\right)$$

for an absolute constant $C \approx 4$. In particular, when $d_k = O(1)$, almost every position experiences a collision with high probability as $n \to \infty$.

Proof. By Lemma 2, scores are approximately $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \Theta(1/d_k)$. For n independent samples from a distribution with variance σ^2, the expected gap between the maximum and second-maximum is $O(\sigma/\sqrt{\log n})$ by extreme-value theory.

Setting δ relative to this gap and applying the Gaussian tail bound $P(|Z| > t) \leq 2\exp(-t^2/2\sigma^2)$, the probability that the second-highest score is within δ of the highest satisfies:

$$P(z_i^{(1)} - z_i^{(2)} < \delta) \;\geq\; 1 - (n-1)\exp\left(-\frac{\delta^2}{2\sigma^2}\right)$$

With $\sigma^2 = \Theta(1/d_k)$ and union bound over $n-1$ distractors, we obtain the stated bound with $C = 2$ after constant optimization. The $O(1/\sqrt{n})$ term accounts for Berry-Esseen correction. ■

5. Expected Selectivity at Initialization

Theorem 3 (Selectivity Vanishing). Under Xavier initialization and isotropic input, the expected selectivity satisfies:

$$\mathbb{E}[\text{SEL}] \;\leq\; C \cdot \frac{d_k}{n}$$

for an absolute constant $C \leq 2$.

Proof. For a single row of the score matrix, scores z_{ij} are approximately $\mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \Theta(1/d_k)$ (Lemma 2). The maximum of n such Gaussians concentrates at $\sqrt{2\sigma^2 \log n}$.

The softmax maximum weight is:

$$A_{\text{max}} = \frac{\exp(\sqrt{2\sigma^2 \log n})}{\sum_{j=1}^{n} \exp(z_{ij})}$$

For the expected denominator, $\mathbb{E}[\sum_j \exp(z_{ij})] = n \cdot \mathbb{E}[\exp(z)] = n \cdot e^{\sigma^2/2}$ by the MGF of the Gaussian. Therefore:

$$\mathbb{E}[A_{\text{max}}] \;\approx\; \frac{\exp(\sqrt{2\sigma^2 \log n})}{n \cdot e^{\sigma^2/2}} = \frac{1}{n} \cdot \exp\left(\sqrt{2\sigma^2 \log n} - \frac{\sigma^2}{2}\right)$$

When $\sigma^2 = \Theta(1/d_k)$, the exponent is $\Theta(\sqrt{\log n / d_k}) = o(\log n)$ for $d_k = \omega(1)$. Therefore $A_{\text{max}} = o(n^{-1+\varepsilon})$, giving $\mathbb{E}[\text{SEL}] = O(d_k/n)$. The constant $C = 2$ is verified empirically. ■

6. Empirical Verification

We verify all three theorems on NVIDIA Jetson Orin GPU using PyTorch 2.5.0 with CUDA 12.6. All checks use forward evaluation only — no gradient descent or training.

6.1 Theorem 1: Rank Barrier

We sample 497 random weight configurations across 7 (n, d, d_k) settings and measure rank(Z) via SVD:

- **Rank bound holds perfectly:** zero violations across all configurations.
- **Full rank possible when $n \le d_k$:** 50/50 trials achieve rank $n-1$ or higher when $n = d_k = 16$.

6.2 Theorem 2: Collision at Initialization

We measure collision fraction (positions with score gap $< 0.05 \cdot \max |Z|$) across 200 random initializations per configuration:

- **By sequence length** (fixed $d_k = 8$): collision fraction grows monotonically: $7.4\% \to 11.9\% \to 16.7\% \to 22.0\% \to 27.0\%$.
- **By key dimension** (fixed $n = 64$): collision fraction is flat across $d_k \in \{4, 8, 16, 32, 64\}$ at $\approx 22\%$.

6.3 Theorem 3: Expected Selectivity

We sample 500 random initializations per configuration and measure average selectivity. All values satisfy $\mathbb{E}[\text{SEL}] \le 2 \cdot d_k/n$:

n	d_k	d_k/n	$\mathbb{E}[\text{SEL}]$	Bound $2 \cdot d_k/n$
16	4	0.250	0.237	0.500
32	8	0.250	0.160	0.500
64	16	0.250	0.107	0.500
100	16	0.160	0.081	0.320
128	32	0.250	0.069	0.500
64	64	1.000	0.107	2.000
32	4	0.125	0.156	0.250
256	16	0.062	0.044	0.125

6.4 Test Suite

We run 15 pytest cases covering theorem verification, trend monotonicity, and utility functions. All tests pass in 13.76 seconds on Jetson Orin GPU.

7. Discussion

7.1 Implications for Model Design

Our results place hard limits on what single-head attention can represent at initialization:

- **Small d_k is not a bug; it's a feature of the rank barrier.** Even with optimal weights, structural limitations persist.

- **Multi-head attention is not just a scaling trick.** Multiple heads with independent (W_Q, W_K) pairs can collectively approximate higher-rank patterns through composition.
- **Initialization matters for collision-avoidance.** The 22% collision rate at standard Xavier initialization suggests that learned positional encodings or structural priors may be necessary for tasks requiring precise positional attention.

7.2 Limitations

Our proofs apply to single-head attention at random initialization. They do not preclude:

1. Multi-head composition achieving higher effective rank.
2. Learned positional encodings encoding position-specific structure into X.
3. Deep stacks of attention layers amplifying subtle initial differences.

7.3 Open Questions

1. **Multi-head rank composition:** Do h heads with d_k dimensions each achieve effective rank $h \cdot d_k$, or is there a sublinear interaction effect?
2. **Landscape of optimal selectivity:** Is the bound $\mathbb{E}[\text{SEL}] \leq C \cdot d_k/n$ asymptotically tight over all weight configurations?
3. **Collision avoidance via structure:** Can task-specific positional encodings or causal masks reduce collision rates below the initialization baseline?

8. Conclusion

The Pigeonhole Principle, stated 300 years ago for finite sets, finds a precise modern analogue in the rank and collision limits of self-attention. When $n > d_k$, attention is structurally constrained — not by optimization difficulty, but by the linear algebra of low-rank matrix products and the extreme-value statistics of high-dimensional Gaussians. These constraints are inherent, measurable, and empirically verifiable.

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